NBA Game-Impact Intelligence

Courtside: Live In-Game Odds and an Agentic Trade Backtest

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DASC7606C, Group 12 Track 2: Agentic Framework Design

Code and report: `https://github.com/Michaelyzr/nba-agent/`

1 Introduction and Problem Definition

1.1 Background

Prediction markets price contracts on future events, including NBA outcomes, as implied probabilities: a useful benchmark, not a guarantee [1]. Polymarket generally shows a bid–ask midpoint rather than an executable price, and a fill depends on the order book [2].

In 2025, state-regulated U.S. sportsbooks took $166.94 billion in wagers and recorded $16.96 billion in gross gaming revenue: the operators' hold and, before promotional credits, customers' aggregate loss to the book [3]. Combined Kalshi and Polymarket monthly notional volume (each contract at its $1 payout) rose from $0.8 billion in July 2024 to $53 billion in July 2026, with sports the largest category on both platforms [4].

1.2 Problem Statement and Target User

The problem is fragmented NBA news set against those prices. A headline can name an injury or lineup change without sizing it, and a forecast does not say whether the price offers value after spreads and fees. Users need a traceable link from that news to the forecast and to a decision to act or pass.

Users include fantasy and sports-media platforms seeking auditable updates, plus broadcasters, writers, team analysts and retail participants. Courtside is the product they open [5]. On a live game it is an odds desk for in-game trading: the price, the model, and a decision the user confirms. The same product backtests that decision with an agent on recorded prices. Fills in both modes stay on the desk; the product does not send an exchange order.

1.3 Project Objective

Courtside connects news, forecasts and prices in one product. The live desk follows a game into the quarters and prices an in-game trade. The agentic framework is the backtester for that trade: it writes a cited decision, and code owns the numbers, the fees and the risk limits. A learning rule is reused only after a historical gate. The test is whether that backtest beats model-only, no-learning and never-trade baselines on recorded Kalshi prices.

The project's thesis is that an agent can make the relationship between evidence, forecasts and prices explicit and auditable. The experiments do not establish a reliable trading edge after costs. Whether the interface improves research speed or user understanding remains a question for user evaluation.

2 Agentic Framework

Courtside is one product with two jobs (Table 1). The live desk is for in-game trading: it carries the odds, the score and the model's win chance from before tip through the quarters, and it turns a selected price into a trade the user can confirm. The agentic framework is the backtester for those trades. It connects a news item to a fee-adjusted edge and records why the decision is to pass. That backtest is not a claim that the agent beats the sportsbook hold.

The backtester is a stateful LangGraph loop with a pregame *decide* phase and a post-settlement *review* phase [6, 7]. The design is market-graded: code owns prices, fees, fills and risk limits, and a learned rule is reused only if it improves held-out closing-line value. LLMs are used where judgement helps—reading news, choosing tools, or arguing that an apparent edge is already priced in.

Two agents share the replay engine and the same caps. `MarketAgent` is the LangGraph control plane. `ToolAgent` replaces the deterministic analyse/propose step with a tool-using analyst and an optional sceptic, behind the same `policy(view,game,now)` interface [10]. The notebook and split gate run in code and do not need an LLM. Backtest decisions fire on news within six hours of tip-off, or at $T - 60$ minutes, once a quote is at most five minutes old. The live desk then continues the game: the in-game price and win chance update with the score and the clock, which is the intra-game half of the product.

Table 1: Courtside as a live in-game odds product and as a trade backtester.

Product mode	What the user gets
Live odds, in-game trading	Demo and Live data. The match page follows the game through the quarters: score, clock, news, model win chance, and the live reference price. Conservative, Balanced and Aggressive set how strong an in-game edge must be. The user confirms the stake on the slip.
Agentic backtest	`MarketAgent` and `ToolAgent` on frozen Kalshi prices, with the sceptic and the learning gate. This mode produces the walk-forward P&L in Section 5.2. Trading tests is that backtest on the desk: the archived season results, and one paper fill from the same edge rule, gap, probability, ask and stake.

2.1 Agent Components and Responsibilities

Table 2 separates who interprets evidence, who estimates probabilities, and who is allowed to act. Analysis adds the M4 news shift in Equation 1; blocked and no-action paths still end at `deliver` with a logged reason. After each market day the review phase is `settle`→`review`→`gate`→(`save_rule`|`reject_rule`).

Table 2: Research-loop components and their implementation boundaries.

Component	Responsibility and boundary
Trigger / investigator	Filter evidence to the decision timestamp; identify affected players using optional LLM interpretation or offline status rules.
Forecast core	Obtain participation, minutes/points distributions and win estimates from `forecast/api.py`, before and after news.
ToolAgent analyst	Call as-of tools (at most eight calls over four turns) and propose buy or pass as JSON. It cannot invent prices.
Market analyst	Add the news shift to an earlier market anchor and compute cost-adjusted gaps. Code, not the LLM, decides whether the edge clears 4¢.
Sceptic	Optional one-sided LLM veto after a validated positive-edge buy. A bad or missing verdict is treated as a reject.
Checks / risk / delivery	Validate citations and numbers; enforce limits and channel permissions. Require retail confirmation; log decisions and blocks.
Review / gate / notebook	Grade settled decisions, propose a restricted rule, test it on earlier days and save only accepted rules.

2.2 Advantages

Replayability and controlled interpretation. Timestamped evidence, forecasts and applied rules are recorded in traces and decision logs. The as-of contract supports comparisons on the same available information, subject to the timing assumptions in Section 3. Numerical checks constrain LLM errors; passing tests does not establish complete robustness.

Bounded tool use. `ToolAgent` may call as-of tools—live quote, 24-hour anchor, public news, M4 win probability and edge helpers—at most eight times across four turns, then emit a JSON proposal (side, probability, citations). Code checks the schema, whether citations match the tools, and whether the stated probability is within three points of a tool-derived reference. Invalid output becomes a pass. Gemini and DeepSeek backends are cached so

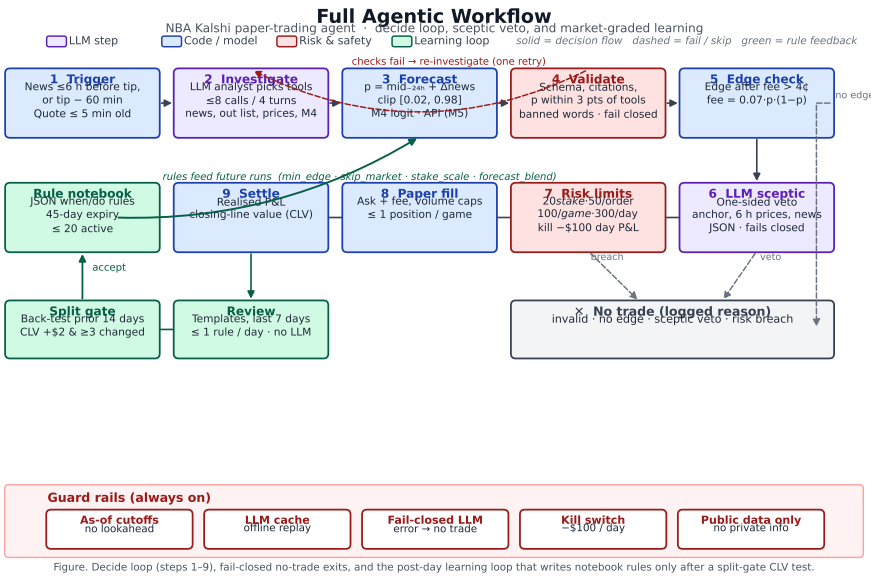


Figure. Decide loop (steps 1–9), fail-closed no-trade exits, and the post-day learning loop that writes notebook rules only after a split-gate CLV test.

Figure 1: Courtside's backtesting decide loop (MarketAgent / ToolAgent): sceptic veto, fail paths, and the learning loop. The live desk is the in-game odds product that this loop is tested against.

a backtest can be replayed. The live odds desk can show the game without making that call [10].

The sceptic. After a buy has already cleared validation and the fee-adjusted edge rule, an optional sceptic sees a fixed bundle: the proposal, the code-computed edge, the 24-hour anchor and the move since then, six hours of prices, the news, and the analyst's tool results. Its prompt asks whether the market already reflects that information. It returns approve or reject. Parse errors and banned wording count as a reject, and the reason is kept short. The sceptic cannot set a new probability or open a trade the code did not already allow. On the labelled set it kept 8 of 182 proposals (Dec–Apr CLV $+7.38) [10]. Later redraws were weaker and less stable, so that figure is a filter result, not a claim of durable edge.

Accountable adaptation. Nightly proposals cannot activate themselves. Restricted rule vocabularies, historical gates and expiry constrain adaptation; split and placebo audits test whether gains reflect better decisions or less exposure [9, 8]. The gate designs appear in Section 3.4; empirical gains and uncertainty appear in Section 5.1.

Live trading and a logged backtest. Media and team channels receive briefs. On the live desk a retail user confirms the in-game stake. In the backtest a pass is a logged outcome, and the season replay stops new pregame entries at tip-off. After tip-off the live desk keeps the in-game price and win chance moving with the game.

3 Data and Methodology

3.1 Data Sources

Experiments use locally frozen, schema-checked Parquet tables, joining ESPN schedules and box scores to Kalshi game-winner contracts, prices and settlements. Table 3 summarizes the frozen inventory [12]. The 3.9 million price rows are repeated observations; the effective win-model sample is games, not quote bars.

Two settled contracts lack matching ESPN games and are excluded from matched trading evaluation. Market candles cover the 30 hours before contract close; each candle becomes visible only at its minute-end timestamp. ESPN provides three seasons of statistics as a practical fallback when NBA endpoints are inaccessible. Historical news uses inactive lists with a reconstructed $T - 30$-minute timestamp, not an observed publication time. This proxy cannot establish actual news-arrival latency or a universal direction of bias.

Live adapters separately retrieve official injury reports, ESPN/CBS feeds and configured Shams, team and reporter accounts, recording provenance and applying source priority. Polymarket Gamma/CLOB clients provide read-only demo references. Historical Polymarket trades are not a recorded order book; any reconstructed spread is an

Table 3: Frozen data inventory and coverage.

Table / source	Coverage and content	Recorded volume
Games / ESPN	Oct 2023–Jun 2026; schedule and final scores	3,962 games
Player-games / ESPN	Minutes, points, attempts and starts	85,422 rows; 811 players
News / ESPN inactive	Oct 2023–May 2026; availability and reasons	3,371 items; 1,772 games
Markets / Kalshi	Oct 2025–Jun 2026; NBA game winners	2,632 contracts; 1,316 games
Prices / Kalshi	One-minute bid, ask and volume observations	3,938,368 quote rows
Settlements / Kalshi	Binary contract outcomes	2,634 contracts

assumption [17, 14].

3.2 Data Preprocessing and Feature Engineering

Seven research tables share schemas, identifiers and UTC timestamps. The replay's read-only `AsOf` view hides unfinished outcomes, unpublished news and future candles. Quotes used in a decision are at most five minutes old. Development precedes 1 February 2026. The matched market test is 501 games through 12 April, and 87 later priced games are the holdout. Training absences come from box scores, while a live override can use only a report that has already been published [12, 11].

3.3 Prediction Models

M1 is the rolling and gradient-boosting baseline. M2 is a GRU for points and minutes. M3 is logistic regression for whether a player plays. M4, the model on the trading path, is logistic regression on six home-minus-away features: weighted margin, recent margin, rest, back-to-backs, missing minutes and missing points. M5 is the shared forecast interface. A neural M4 and a quote-sequence M6 were tried and did not beat these simpler models, or a zero-move benchmark, by enough to clear the spread and the fee [12, 13].

The live desk's in-game win chance is a diffusion update ($\sigma = 14$) from the pre-tip probability, the score and the clock [14]. That update was not fitted on the season sample, and the backtest in Section 5.2 uses the pregame anchor and the M4 news shift.

3.4 Trading Strategy

At each pregame trigger the research policy anchors to the midpoint 24 hours before tip (or its earliest available quote), adding M4's news shift:

$$\widehat{p} = \mathrm{clip}_{[0.02,0.98]}\left(q_{-24h} + p_{\mathrm{M4,after}} - p_{\mathrm{M4,before}}\right). \tag{1}$$

For YES, $p_s = \widehat{p}$ and $c_s =$ ask; for NO, $p_s = 1-\widehat{p}$ and $c_s = 1-$bid. A candidate requires $p_s - c_s - 0.07c_s(1-c_s) > 0.04$. Screening uses the continuous fee approximation; settlement rounds the aggregate fill fee up to cents. Default stakes are \$20; optional quarter-Kelly sizing uses a fixed \$1,000 bankroll. Only one position is held per game. Fresh pre-tip fills are limited to 10% of recent volume (or a 200-contract fallback when volume is missing).

Code enforces \$50/order, \$100/game and \$300/day caps, citations and channel permissions. A \$100 realized-loss stop is configurable; headline reproduction runs did not enforce it. The nightly reviewer proposes at most one restricted skip/edge/stake rule. Legacy gates require 0.005 mean-CLV improvement on at least three changed trades; split gates separate seven selection days from 14 earlier evaluation days and require \$2 total-CLV improvement. Accepted rules expire after 45 days [8].

Settlement records fee-adjusted P&L and CLV (closing midpoint minus entry). Brier/log loss grade probabilities; pinball loss and coverage grade quantiles. Intervals resample game-days, using 2,000 replicates for the reported model comparisons. Section 5.4 is the same product in the game: a five-point in-game edge and one contract. The season figures are the backtest; the live desk is where the user confirms an in-game trade.

4 Implementation and User Workflow

Courtside is the product. Start it with `python -m agents.dashboard_server`. Three tabs open: Demo, Live data, and Trading tests. Demo and Live data are the in-game odds desk. Trading tests is the agentic backtest of those trades.

Live odds for in-game trading. Demo and Live data share the match page. A pregame controller and an in-play controller keep their own state, so the page can follow one game from the listing through tip-off and the quarters. The latest valid pre-tip probability starts the live view. After tip-off the win chance updates from the score, the clock and availability, and the page sets that chance against the live reference price [14]. Fixtures follow the current NBA date in Eastern Time. The page shows fair odds, Polymarket references and news [16, 17, 15]. For probability p, all-in decimal odds D and stake S, the desk computes

$$D_{\text{fair}} = \frac{1}{p}, \qquad p_{\text{break-even}} = \frac{1}{D}, \qquad \mathbb{E}[\text{net profit}] = S(pD - 1). \tag{2}$$

Analysis accounts for depth, estimated fees and a cost buffer. Missing, closed or stale references block assessment.

On the live slip, Conservative, Balanced and Aggressive are the in-game acceptance settings: at least 60% estimated chance and more than four points of price edge, 45% and two points, and 25% and 0.5 points. Negative expected value passes none. The user keeps the stake. Parlays are a slip feature, not part of the backtest. The user confirms the trade on the desk.

Agentic backtest. Trading tests is the third tab, and it is the backtester on the same desk. It archives how the agent decides—the decide path, the frozen sceptic (8 of 182, Dec–Apr CLV \$+7.38), the walk-forward P&L, and the later checks—and then runs one paper trade from the published rule. The form takes an edge gap (default 4¢), a stake (default \$20), a home-win probability in $[0.02, 0.98]$, and an ask. The edge is $p - \text{ask} - 0.07\,\text{ask}(1 - \text{ask})$. The desk shows a paper fill when that edge, in cents, is strictly above the gap, and a pass otherwise. A fill is clamped at \$50. The \$100/game and \$300/day caps are part of the backtest policy and are shown on the tab. This run uses the published edge rule on the numbers the user enters, so a single form does not place a season replay or a live sceptic call.

5 Experimental Evaluation and Results

5.1 Prediction Model Performance

On held-out games, M2, M3 and M4 each beat their simple baseline: gradient boosting for points and minutes, last-game playing status, and M4 without absence features. That is not the comparison that matters for a trade. On 501 matched games from 1 February to 12 April 2026, the Kalshi midpoint at tip has Brier score 0.1633. M4 with absences known is 0.1827, worse than the market one hour before tip by 0.0186 (95% CI 0.0107–0.0271). Adding the M4 news shift to the 24-hour market moves that anchor from 0.1676 to 0.1663, which does not beat the contemporaneous price [12]. The backtest therefore treats the market as the anchor and M4 as a news adjustment, not as a forecast that replaces the book.

5.2 Agent trading results

These dollars are Courtside's agentic backtest on frozen prices. Trading tests is that backtest on the desk: it shows this comparison and can run one paper fill from the same rule. The comparison uses the same recorded prices, the same \$20 stake and the same caps. Never trading is \$0. The full agent anchors on the 24-hour market, adds the M4 news shift, and applies the fee-adjusted 4¢ rule plus the learning gate. The no-learning agent keeps that anchor and edge rule with an empty notebook. The raw model trades M4 with no market anchor and no agent. Table 4 is the monthly walk-forward from November through 12 April [11].

Every policy that trades loses money after fees, and every one has negative closing-line value, so none beats the closing price. The dollar loss shrinks as the policy trades less. On the same game-days the full agent finishes \$382 ahead of the no-learning agent (95% CI −\$304 to +\$1,047) and \$2,450 ahead of the raw model (95% CI +\$854

Table 4: Walk-forward P&L after fees, November–12 April. Intervals are 95% game-day bootstrap CIs.

Policy	Trades	P&L after fees	ROI	CLV $
Full agent (anchor + learning)	157	−$25 [−569, +564]	−0.8%	−44
Agent, anchor, no learning	276	−$407 [−1,324, +553]	−7.5%	−84
Raw model, no agent	711	−$2,475 [−3,995, −845]	−17.6%	−255
Never trade	0	$0	–	0

to +$3,984). That first gap is the paired learning gain: both agents see the same days, and the figure is the sum of the daily P&L differences. Mean CLV per trade does not move when learning is on (difference −0.0002). The extra dollars come from skipping trades, not from a better fill on the trades that remain. Against never trading, the full agent's P&L interval still covers zero, while its CLV of −$44 does not.

The February–April split used for the original ablations ranks the policies the same way. The full agent loses $32 on 74 trades. Turning learning off, or freezing the development rules, loses $247 on 129 trades. Dropping the market anchor, which is also what a plain model does with no agent, loses $2,087 on 366 trades. A win-rate placeholder loses $429 on 246 trades. In the later 87-game holdout the full agent takes no trades. The no-learning agent loses $74 on six trades. The raw model shows +$450 after fees (95% CI −$55 to +$1,171) with negative CLV, so that profit is not a close-beating result. The anchor and the reviewer were designed after early test inspection; walk-forward checks reduce training leakage but cannot undo that design history.

The LLM policies were scored on a fixed 40% subsample of the February–April decision points, so their dollar totals are not on the same scale as Table 4 [10]. On that subsample the deterministic anchor loses $236 (53 trades). With DeepSeek thinking on, a one-call LLM makes five trades for +$4; the tool agent without a sceptic makes 42 trades and loses $93; with the sceptic it keeps two trades for +$100 (95% CI −$62 to +$363). The chat model with the sceptic keeps no trades, which is the never-trade result. Those intervals cover zero, or rest on two trades. Results are from saved artifacts, not rerun experiments.

5.3 Attempts to Improve the Agentic Framework

Several checks were run after the submission freeze as robustness, not as a replacement for the frozen headline. On that headline the sceptic kept 8 of 182 validated proposals (Dec–Apr CLV $+7.38; four paper trades in the Feb–Apr window). The walk-forward comparison above is unchanged. Table 5 records the later attempts; none created a reliable edge.

Table 5: Post-freeze attempts to improve the agent, and their outcomes.

Attempt	Outcome
Sceptic redraws and a softer prompt	Mean CLV about $+3.27, Fleiss $\kappa \approx 0.15$. A balanced prompt had AUC ≈ 0.53. The published $+7.38 was an unstable draw.
Majority of five calls	Agreement rose ($\kappa \approx 0.39$), but only two trades were kept (CLV $+0.83). None of the original eight approvals survived.
LLM win probability	Correlation with the live mid ≈ 1: the model echoed the price.
Other filters	A learned filter, a 1–2¢ pre-filter, and using M4 as the anchor instead of the market all failed or were left off. Injury timing was about −1.4¢ and trade timing about −2¢ after costs.
Perfect CLV filter	An after-the-fact ceiling on the same 182 proposals: 55 trades, about $+39.87 (~3.6% on stakes).

The perfect-filter row is an upper bound on redesigning the sceptic for this market, not a rule that can be traded: it keeps winners after the fact. These checks support gated abstention, not a durable edge on NBA game-winner contracts after fees.

5.4 Case Study: In-Play Market Comparison and Trading Simulation

The in-game half of Courtside is shown on Sacramento at Orlando, 26 March 2026 (final 117–121). Figure 2 follows the live-desk win chance against the Kalshi book and Polymarket trades from the scheduled tip. At Q4 08:49 and a 95–95 tie, that chance is 64.2% for Orlando, against trade prices of 0.7600 and 0.7675. Each in-game strategy buys one contract at its first five-point edge and holds to settlement [19].

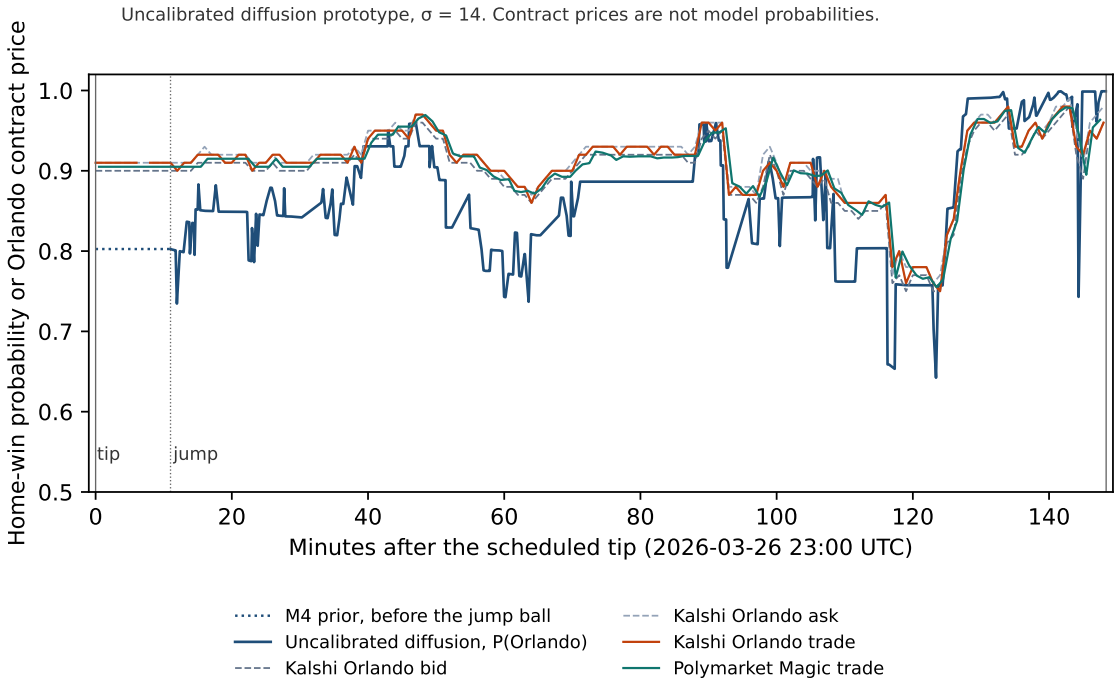


Figure 2: Live in-game trading on Orlando, 26 March 2026. The diffusion update ($\sigma = 14$) is the desk's win chance. Kalshi bid, ask and trades, and Polymarket trades, are the live odds. Contract prices are not model probabilities.

Table 6: Single-game simulation, after the assumed fee.

Strategy	Position	Entry quote	Net P&L	Net ROI
YES-only	Orlando YES	\$0.84	+\$0.15	+17.65%
NO-only	Orlando NO	\$0.10	−\$0.11	−100%
Two-sided	Orlando NO	\$0.10	−\$0.11	−100%

The two-sided policy enters NO because its prior sits below the market, and its one-trade limit blocks a later YES. This is one in-game trading path on the live desk, settled on the recorded prices [19].

6 Limitations

The replay stamps ESPN inactive lists 30 minutes before tip, and on 193 test games most of the price move in the M4-news direction is already in the book by then [20, 21]. Box-score absences are not prospective reports. The market beats the model on matched games, and spreads, fees and delay can erase an apparent edge. Learning cuts the number of trades rather than the closing-line value of each one. There is no durable edge after costs, and the single-game chart is one recorded path, not an executable return. A user study of whether the desk is faster to read is still open.

LLM usage

The ToolAgent and sceptic sections use Gemini and DeepSeek to propose trades and to veto an apparent edge. Code validates the schema, the citations and the probability, and fail-closes: a bad reply becomes a pass. An LLM drafted and edited the prose of this report, including grammar, across the report. All prices, fills, caps and statistics come from saved evaluation artifacts, not from the model.

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